

Rainwater infiltration and soil water retention

Spatial scale: vineyard

About this sustainable management strategy

Rainwater infiltration and soil water retention are nature-based practices that help keep rainfall within the vineyard by slowing runoff and allowing water to infiltrate into the soil. Instead of directing water away from the field, these practices use natural soil and vegetation processes to store moisture where vines can access it during dry periods.

The strategy can be implemented by creating or maintaining shallow vegetated depressions, infiltration strips, or contour-aligned swales between vineyard rows that temporarily collect rainwater after storm events. These features should be vegetated with permanent ground cover and combined with practices such as mulching, compost application, or reduced tillage to improve soil structure and increase the soil's capacity to absorb and retain water.

Regular maintenance involves removing sediment that may reduce infiltration, maintaining vegetation cover, and checking that water drains gradually after rainfall. The design and placement of infiltration areas should follow the natural slope of the vineyard and be adapted to local soil conditions, rainfall patterns, and machinery traffic to avoid interference with routine vineyard operations.

Key ecosystem services:

- Water retention (*reduced surface runoff*)
- Water regulation (*enhanced infiltration and groundwater recharge*)
- Flood and erosion control
- Soil health and fertility (*improved soil structure, organic matter and nutrient cycling*)
- Biodiversity enhancement (*above- and below-ground*)
- Micro-climate regulation (*greater resilience to drought and extreme rainfall*)

Considerations for successful implementation

Adapting the design to site conditions

The size, depth, and location of infiltration features should reflect vineyard slope, soil type, and expected rainfall. On steeper slopes, several small infiltration areas are generally more effective than a single large one.

Maintaining vegetation cover

Permanent vegetation within infiltration areas protects the soil surface, slows water movement, filters sediments, and supports beneficial organisms. Bare soil should be avoided wherever possible.

Preventing clogging and waterlogging

Sediment, pruning residues, or compacted soil can reduce infiltration over time. Regular inspection after heavy rainfall helps identify areas that require cleaning or light soil loosening. Water should infiltrate within a reasonable period and not remain standing long enough to damage vine roots.